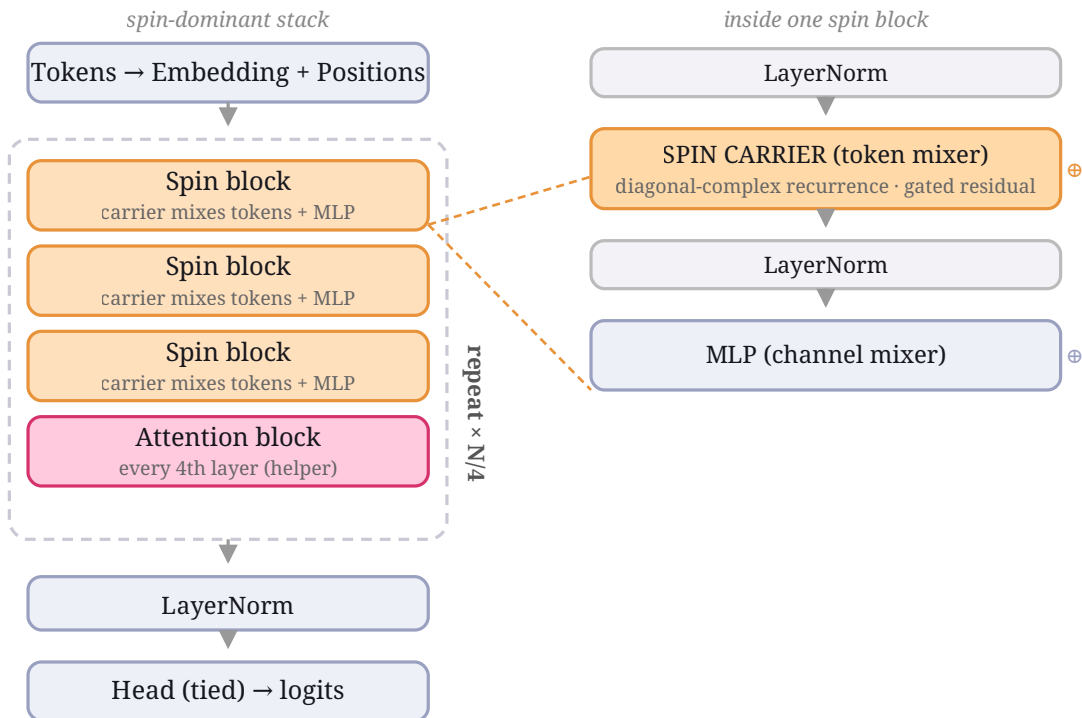




The spin carrier

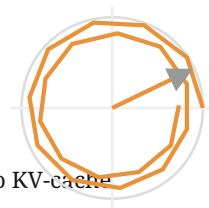
$$h_t = \lambda \odot h_{(t-1)} + b_t$$

$$\lambda_j = \exp(-\exp v_j) \cdot \exp(i \theta_j)$$

a damped rotation per channel: θ_j sets the turn, $|\lambda_j| < 1$ the decay

Run as a **log-depth parallel associative scan** -- not a T-step loop.

Cost $O(T \cdot d)$ vs attention $O(T^2 \cdot d)$; at inference recurrent, $O(1)$ /token, no KV-cache



damped rotation (the "spin")